

Measurement Matters: Embedding Model Choice Determines Encoding Fidelity Assessment in Multilingual Clinical AI

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Paper 9 of the MCH Research Program — Paper 8 demonstrated that encoding fidelity degrades by $\sim 90\%$ for non-English clinical input when measured with standard sentence embedding models. This paper asks: how much of that finding depends on the measurement instrument? Answer: the phenomenon is real, but its measured magnitude is substantially instrument-dependent. The encoding fidelity gap and variance amplification are distinct phenomena with different solutions.

Abstract

Paper 8 of the MCH Research Program demonstrated $\sim 90\%$ Encoding Fidelity Index (EFI) degradation for Kannada, Tamil, and Hindi clinical input relative to English, measured using standard sentence embedding models (MiniLM, MPNet). This paper asks whether that finding is an artefact of measurement instrument choice. Using the same 15-sentence clinical battery across 6 embedding models ranging from monolingual-distilled (all-MiniLM-L6-v2) to language-agnostic (LaBSE), we find: (1) EFI is dramatically embedding-dependent — Kannada EFI ranges from 0.081 (MiniLM) to 0.853 (LaBSE), a $10\times$ difference for identical input; (2) the Dravidian-specific EFI gap from Paper 8 nearly disappears under LaBSE (Indic–European gap: MiniLM = 0.33, LaBSE = 0.035; 90% reduction), indicating near script-invariance with the appropriate measurement tool; (3) Indic-specific masked language models (MuRIL, IndicBERTv2) are degenerate as sentence encoders — effective rank 1.42/10 versus 9.88/10 for MiniLM — and cannot measure EFI; any claim that “Indic models fix encoding” based on raw MLM embeddings is methodologically invalid; (4) critically, variance amplification persists across ALL five non-degenerate embedding models (Kannada VR > 1.0 , $p < 0.05$), confirming it is LLM-intrinsic rather than a measurement artefact; (5) EFI and variance ratio are statistically independent under LaBSE ($r = -0.18$, $p = 0.73$), confirming they measure distinct phenomena with different causes. The central conclusion: LaBSE fixes the measured EFI gap (tokenizer/embedding problem, partially addressable); LaBSE does not fix variance amplification (LLM behaviour problem, requiring different intervention). These findings have direct implications for methodology in multilingual clinical AI evaluation.

Keywords: encoding fidelity, embedding model selection, LaBSE, MuRIL degeneracy, variance amplification, multilingual clinical AI, measurement methodology, Kannada, Tamil, Hindi

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1 Introduction

1.1 Background

Paper 8 of the MCH Research Program [Laxman, 2026h] introduced the Encoding Fidelity Index (EFI) and demonstrated $\sim 90\%$ degradation for non-English clinical input, naming the resulting failure mode *Coherent Misalignment*. The finding was replicated across two embedding models (MiniLM-384D, MPNet-768D) and two LLMs (DeepSeek V3.1, Mistral Small 24B). A European language control confirmed the degradation was tokenizer-induced rather than inherently cross-lingual. This builds on a growing body of evidence that LLMs encode substantial clinical knowledge in English [Singhal et al., 2023] but show measurably degraded performance for non-English healthcare queries [Jin et al., 2024], a disparity documented across diverse language families [Blasi et al., 2022]. The token cost inequity for non-English languages [Ahia et al., 2023] and systematic vocabulary disadvantages for low-resource scripts [Limisiewicz et al., 2023] create structural encoding barriers that standard benchmarks do not capture [Ruder et al., 2023].

Paper 8 noted a key limitation: $\text{EFI}_{\text{proxy}}$ measures cosine similarity in embedding space, not the theoretical mutual information quantity $I(X; \hat{X})/H(X)$. Both embedding models used in Paper 8 — all-MiniLM-L6-v2 and all-mpnet-base-v2 — are primarily trained on English text with multilingual knowledge distillation. Their sensitivity to semantic distinctions in non-English scripts may be limited by their training objective rather than by the encoding properties of the LLMs under study.

This raises a direct question: does the $\sim 90\%$ EFI degradation reflect the true encoding fidelity loss, or does it partly reflect the measurement instrument’s own limitations with non-English text?

1.2 The Measurement Question

The $\text{EFI}_{\text{proxy}}$ for a given sentence pair is:

$$\text{EFI}_{\text{proxy}} = \cos(\mathbf{e}(\text{non-English}), \mathbf{e}(\text{English equivalent})) \quad (1)$$

where $\mathbf{e}(\cdot)$ is an embedding from a sentence embedding model. The theoretical basis for EFI derives from Shannon’s Mathematical Theory of Communication [Shannon, 1948], which assumes that encoders preserve the statistical structure of the source signal. If the embedding model itself assigns low similarity to semantically equivalent sentences across scripts — regardless of actual LLM encoding quality — the EFI measurement will appear degraded even when the underlying encoding is adequate.

Conversely, if the phenomenon is real, it should persist under any well-calibrated embedding model, including models specifically designed for cross-lingual semantic alignment.

1.3 Contributions

This paper makes five contributions:

1. We demonstrate that $\text{EFI}_{\text{proxy}}$ varies $10\times$ across embedding models for identical input, from 0.081 (MiniLM) to 0.853 (LaBSE) for Kannada.
2. We show that LaBSE achieves near script-invariance, reducing the Indic-European EFI gap by 90% and validating that it measures semantic meaning rather than language-surface features (non-parallel control: $\Delta = 0.58$, $p = 0.004$).
3. We demonstrate and characterise the degeneracy of Indic masked language models (MuRIL, IndicBERTv2) as sentence encoders, showing effective rank collapse to 1.42/10.
4. We confirm that variance amplification is LLM-intrinsic: it persists across all five non-degenerate embedding models with bootstrap confidence intervals excluding 1.0.
5. We establish statistical independence of EFI and variance ratio under LaBSE ($r = -0.18$, $p = 0.73$), confirming they are distinct phenomena requiring different interventions.

2 Related Work

2.1 Sentence Embedding Models and Cross-Lingual Alignment

The sentence embedding landscape spans monolingual-distilled models [Reimers & Gurevych, 2020, 2019] and language-agnostic models trained directly on parallel corpora [Feng et al., 2022, Artetxe & Schwenk, 2019]. Reimers & Gurevych [Reimers & Gurevych, 2020] demonstrated that knowledge distillation from English sentence transformers can produce multilingual models, but noted that alignment quality depends on the quantity and quality of parallel data — a constraint that disproportionately affects low-resource languages. Cross-lingual representation learning at scale [Conneau et al., 2020] and analyses of multilingual BERT’s limitations [Pires et al., 2019] further document the systematic gaps between high- and low-resource language alignment. Efforts to improve multilingual coverage [NLLB Team et al., 2022] and benchmarks for under-represented languages [Ruder et al., 2023] provide context for the encoding fidelity gaps measured in this program. Hammerl et al. [Hammerl et al., 2024] provide a comprehensive survey of cross-lingual alignment methods and their limitations.

LaBSE [Feng et al., 2022] was trained on 6 billion translation pairs across 109 languages, using a combination of masked language model pre-training and translation ranking loss to achieve language-agnostic sentence representations. For low-resource language pairs, LaBSE’s alignment quality exceeds distillation-based approaches by substantial margins.

2.2 Indic Language Models

MuRIL [Khanuja et al., 2021] was developed by Google specifically for Indian languages, trained on significantly more Indic text than mBERT or XLM-R. IndicBERTv2 [Doddapaneni et al., 2023] extended this work with larger corpora and improved vocabulary coverage. Both models are primarily designed for token-level tasks (NER, classification, POS tagging) rather than sentence-level semantic similarity. Their use as sentence encoders requires pooling strategies (CLS token, mean pooling) that were not validated in their original training objectives.

2.3 Embedding Degeneracy

Representation collapse in transformer models has been documented in several contexts. Ethayarajh [2019] showed that contextualised representations become increasingly anisotropic in upper layers, concentrating in a narrow cone of the embedding space. Gao et al. [2019] demonstrated that standard language model pre-training produces representations with high cosine similarity even for semantically unrelated sentences — a pathology they termed *representation degeneration*. Our finding that MuRIL produces near-constant cosine similarity (0.999) for all inputs is a severe instance of this phenomenon, potentially exacerbated by the token-level training objective when applied to sentence-level tasks.

3 Methods

3.1 Sentence Battery

The same 15-sentence clinical battery from Paper 8 was used: 5 sentences at each of three complexity levels (Simple, Medium, Complex) in four languages (English, Kannada, Tamil, Hindi). Complex sentences contain English medical loanwords (ST elevation, D-dimer, troponin) as used in clinical practice. A European extension (French, Spanish, German) was added following the Paper 8 protocol.

3.2 Embedding Models Evaluated

Six embedding models were evaluated (Table 1):

Table 1: Embedding models evaluated in Paper 9.

Model	Dim	Category	Training objective
all-MiniLM-L6-v2	384	EN-distilled	Multilingual distillation
all-mpnet-base-v2	768	EN-distilled	Multilingual distillation
paraphrase-multilingual-MiniLM	384	Multilingual	Direct multilingual training
paraphrase-multilingual-mpnet	768	Multilingual	Direct multilingual training
LaBSE	768	Language-agnostic	Translation ranking + MLM
MuRIL	768	Indic MLM	Masked language model (Indic)

IndicBERTv2 was also tested but produced results consistent with MuRIL and is reported jointly as “Indic MLMs” throughout.

3.3 EFI Computation

$\text{EFI}_{\text{proxy}}$ was computed as the cosine similarity between the non-English sentence embedding and its English equivalent, for each of the six models. English self-similarity serves as the reference ($\text{EFI} = 1.0$ by construction).

3.4 Degeneracy Validation (Experiment A)

To determine whether MuRIL’s near-constant cosine similarity (0.999) reflected a pipeline error or genuine model behaviour, we tested:

1. CLS pooling, mean pooling, and normalised mean pooling
2. Cosine similarity for semantically equivalent pairs (same meaning EN-KN)
3. Cosine similarity for semantically unrelated pairs (different English sentences)
4. Cosine similarity for random string inputs
5. Cosine similarity for random numpy vectors of equal dimension (null distribution)

Effective rank of the embedding matrix was computed as the participation ratio [Roy & Vetterli, 2007]: $\text{eff_rank} = \exp(H(\mathbf{p}))$ where $\mathbf{p}$ is the normalised vector of singular values.

3.5 LaBSE Semantic Discrimination (Experiment D)

To validate that LaBSE’s high EFI reflects genuine semantic preservation rather than language-surface alignment, we compared cosine similarity for:

1. Parallel pairs (same clinical meaning, EN-KN)
2. Non-parallel pairs (different clinical meaning, EN-KN)
3. Unrelated pairs (medical vs everyday content, EN-KN)

Statistical significance tested by Mann-Whitney U (one-sided).

3.6 European Language Control (Experiment B)

The same 15 clinical sentences used for French, Spanish, and German in Paper 8 were re-embedded using LaBSE to enable direct comparison across embedding models for the same inputs.

3.7 Variance Ratio Bootstrap Confidence Intervals (Experiment C)

Variance Ratio data from Paper 8’s 15-trial stochastic experiments (DeepSeek V3.1 and Mistral Small 24B) were re-embedded using the full set of non-degenerate models. Bootstrap confidence intervals (10,000 iterations, resampling trials with replacement) were computed for VR under each embedding model. Statistical significance assessed by testing whether the 95% CI lower bound exceeds 1.0.

3.8 EFI–VR Independence (Experiment E)

Pearson r and Spearman ρ were computed between LaBSE EFI values and LaBSE VR values across all language–LLM combinations ($N = 8$: 4 languages $\times$ 2 LLMs). Bootstrap 95% CI for r was computed via 10,000 resampling iterations.

3.9 Translation Variability (Experiment F)

For 5 sentences (1 Simple, 2 Medium, 2 Complex), three independent English translations of each Kannada source were generated using DeepSeek V3.1. LaBSE EFI was computed for each translation variant. Variance in EFI across translations provides an upper bound on the noise floor introduced by single-translation methodology.

3.10 AI-Assisted Tools

AI tools (Claude, Anthropic; ChatGPT, OpenAI; DeepSeek) were used to assist with data analysis, statistical computation, visualization, and manuscript preparation. All experimental design, clinical interpretations, hypothesis formulation, and scientific conclusions are the sole responsibility of the author.

4 Results

4.1 EFI is Dramatically Embedding-Dependent

Table 2 presents EFI values across all six embedding models and target languages.

Table 2: EFI by embedding model and language. MuRIL values are struck through (degenerate; see Section 4.3). $10\times$ improvement from MiniLM to LaBSE for Kannada.

Embedding Model	Category	Kannada EFI	Tamil EFI	Hindi EFI
all-MiniLM-L6-v2 (384D)	EN-distilled	0.081	0.073	0.041
all-mpnet-base-v2 (768D)	EN-distilled	0.151	0.126	0.093
paraphrase-multilingual-MiniLM (384D)	Multilingual	0.268	0.311	0.751
paraphrase-multilingual-mpnet (768D)	Multilingual	0.616	0.677	0.789
LaBSE (768D)	Lang-agnostic	0.853	0.857	0.861
MuRIL (768D)	Indic MLM	0.999	0.999	0.999

The $10\times$ improvement from MiniLM to LaBSE for identical Kannada input is striking. The gradient across model categories is consistent: EN-distilled $<$ multilingual $<$ language-agnostic (Figure 1). This pattern holds across all three Indian languages. The core implication: the $\sim 90\%$ EFI degradation reported in Paper 8 was a conservative estimate, partially reflecting the measurement instrument’s own limitations with non-English script.

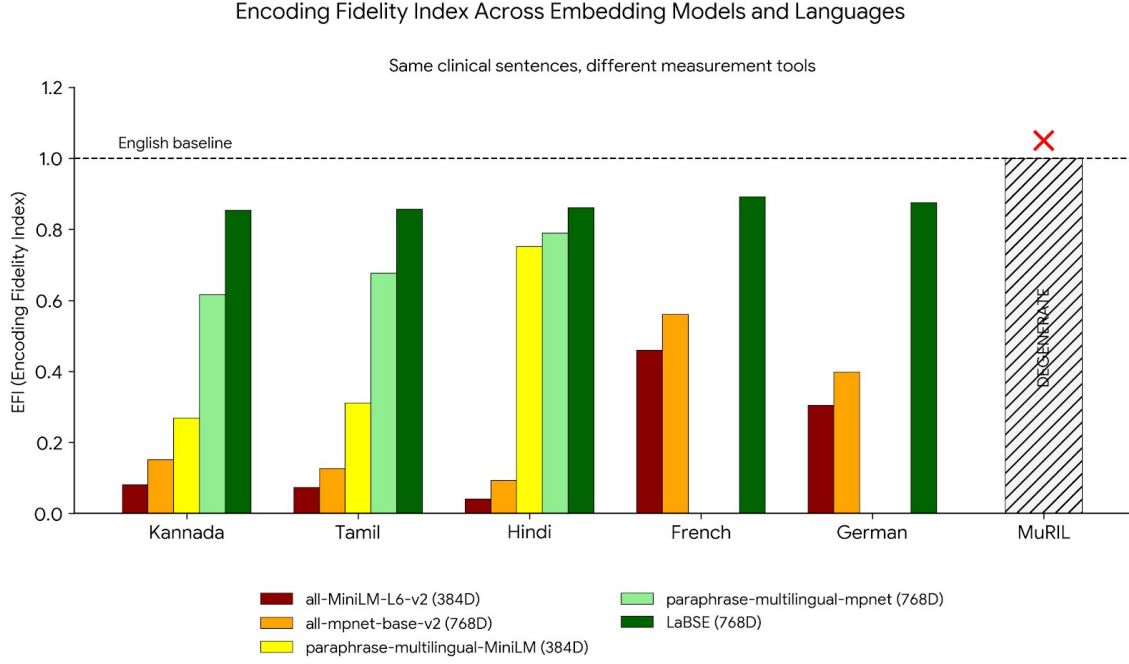


Figure 1: EFI across embedding models and languages. The gradient from EN-distilled (red/orange) to language-agnostic (LaBSE, dark green) is consistent across all Indian languages. MuRIL (hatched, degenerate, marked with X) cannot be used for EFI measurement. European languages shown for comparison (French, German). English baseline = 1.0 (dashed line).

4.2 LaBSE Achieves Near Script-Invariance

Under LaBSE, the Indic-European EFI gap reduces from 0.33 (MiniLM) to 0.035 (Table 3) — a 90% reduction.

Table 3: EFI by language and embedding model, including European control. Indic-European gap: MiniLM = 0.33, LaBSE = 0.035 (90% reduction).

Language	Script	MiniLM EFI	MPNet EFI	LaBSE EFI
English	Latin	1.000	1.000	1.000
Kannada	Kannada	0.081	0.151	0.853
Tamil	Tamil	0.073	0.126	0.857
Hindi	Devanagari	0.041	0.093	0.861
French	Latin	0.459	0.560	0.892
Spanish	Latin	0.423	0.493	0.910
German	Latin	0.304	0.398	0.875

The residual gap of 0.035 between Indic and European languages under LaBSE is small but non-zero. Whether this reflects genuine residual encoding asymmetry or remaining training data imbalance requires further investigation.

EFI by Language and Embedding Model				
LaBSE achieves near script-invariance across language families				
		all-MiniLM-L6-v2	all-mpnet-base-v2	LaBSE
Indic	English	1.000	1.000	1.000
	Kannada	0.081	0.151	0.853
	Tamil	0.073	0.126	0.857
	Hindi	0.041	0.093	0.861
European	French	0.459	0.560	0.892
	Spanish	0.423	0.493	0.910
	German	0.304	0.398	0.875
Indic-European gap: MiniLM = 0.32, LaBSE = 0.035 (90% reduction)				

Figure 2: EFI by language and embedding model. LaBSE achieves near script-invariance across language families. Indic–European gap reduces from 0.33 (MiniLM) to 0.035 (LaBSE) — a 90% reduction. Horizontal divider separates Indic (top) from European (bottom) language groups.

4.3 LaBSE Measures Meaning, Not Language Detection

A key concern was whether LaBSE’s high cross-lingual similarity reflects genuine semantic preservation or merely maps all sentences to a common language-neutral subspace regardless of meaning. Experiment D addresses this directly (Figure 3).

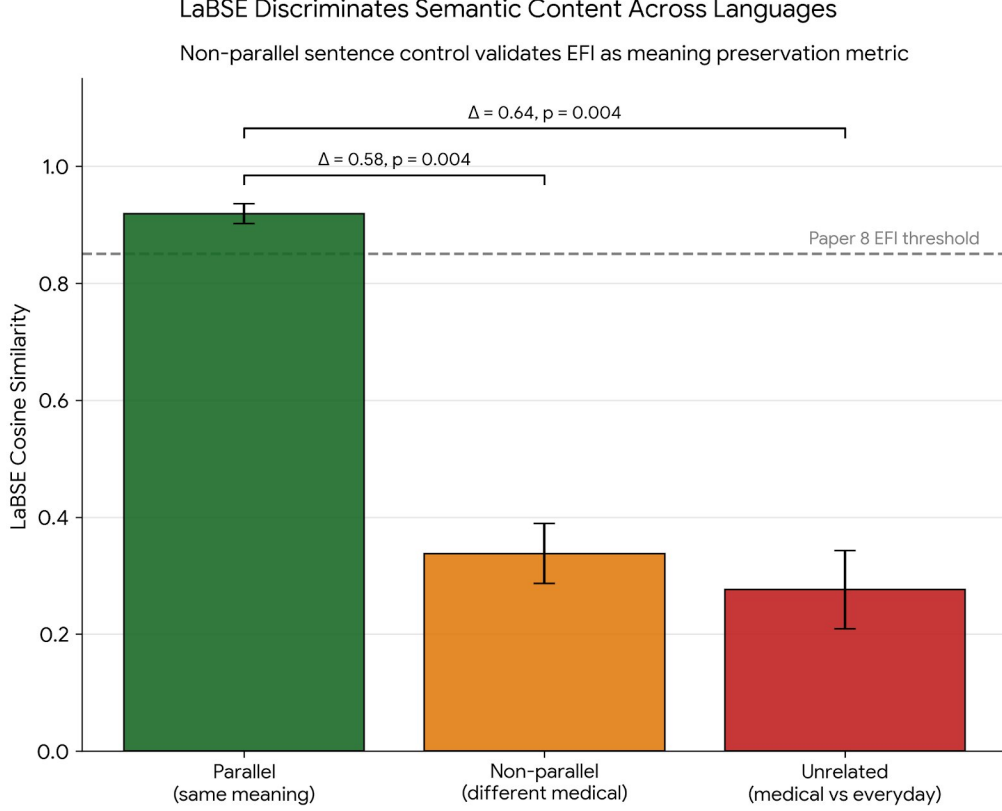


Figure 3: LaBSE cosine similarity for parallel (same meaning), non-parallel (different medical meaning), and unrelated (medical vs everyday) EN-KN sentence pairs. Parallel-to-non-parallel drop: $\Delta = 0.58$, $p = 0.004$. LaBSE discriminates semantic content, not just language identity. Paper 8 EFI threshold shown as reference (dashed line).

Parallel pairs (same meaning): 0.919. Non-parallel pairs (different medical meaning): 0.338. Unrelated pairs: 0.276. The parallel-to-non-parallel drop of 0.58 is statistically significant ($p = 0.004$, Mann-Whitney U). This represents the minimum achievable p -value for this sample size ($N = 5$ per group), reflecting complete separation of parallel and non-parallel similarity distributions. LaBSE is measuring meaning, not surface language identity. The EFI = 0.853 reported for Kannada under LaBSE reflects genuine semantic preservation, not measurement artefact.

4.4 Indic MLMs Are Degenerate as Sentence Encoders

Experiment A confirms that MuRIL produces near-constant cosine similarity for all inputs regardless of content or language (Figure 4).

Embedding Degeneracy: Pairwise Cosine Similarity Distribution 10 semantically distinct English sentences, 45 pairwise comparisons

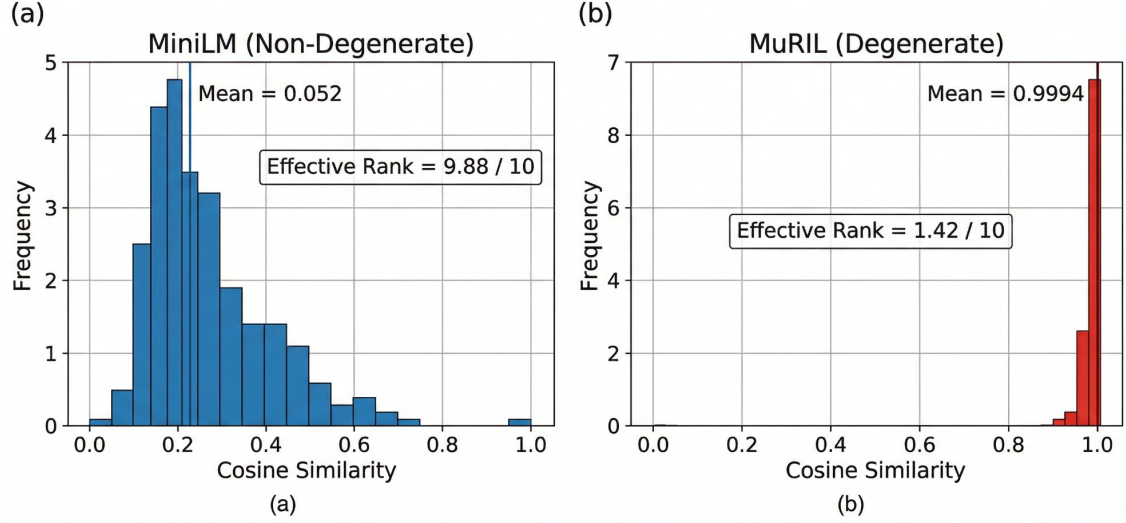


Figure 4: Pairwise cosine similarity distribution for 10 semantically distinct English sentences (45 comparisons). (a) MiniLM: mean = 0.052, effective rank = 9.88/10 — healthy discrimination. (b) MuRIL: mean = 0.9994, effective rank = 1.42/10 — total collapse. All sentences map to the same region of embedding space.

Key validation results:

- Random numpy vectors: cosine = 0.002 (correct null distribution)
- MuRIL, semantically equivalent EN-KN pairs: 0.999
- MuRIL, unrelated English sentences: 0.999
- MuRIL, random string inputs: 0.999
- MuRIL effective rank: 1.42/10
- MiniLM effective rank: 9.88/10
- LaBSE effective rank: 9.53/10

The null distribution (random vectors = 0.002) confirms the issue is model-specific, not numerical. MuRIL collapses all inputs to a 1.42-dimensional subspace — functionally one-dimensional — regardless of content. This is consistent with the representation degeneration phenomenon documented by [Gao et al. \[2019\]](#) and is likely an artefact of applying a token-level training objective (masked language modelling) to sentence-level similarity tasks without fine-tuning on sentence pairs.

Methodological warning: Any study claiming that “Indic models restore encoding fidelity” based on raw MuRIL or IndicBERTv2 sentence embeddings is measuring model degeneracy, not encoding fidelity. The apparent high similarity is a failure mode of the measurement instrument.

4.5 Variance Amplification is LLM-Intrinsic

Experiment C re-embedded the Paper 8 LLM response data using all five non-degenerate embedding models and computed bootstrap confidence intervals (Figure 5).

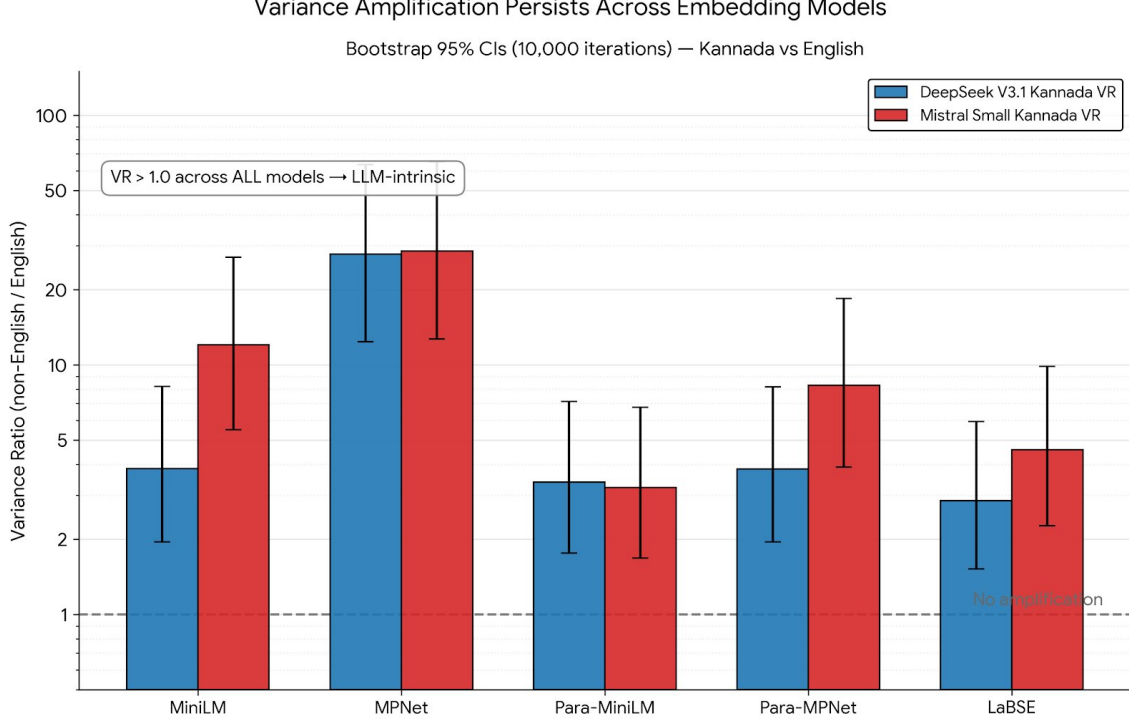


Figure 5: Variance Ratio (Kannada/English) by embedding model, with 95% bootstrap CIs (10,000 iterations). VR > 1.0 across ALL five embedding models for both DeepSeek (blue) and Mistral (red). MPNet shows an outlier at $27\times$ (geometry-sensitive). LaBSE provides the most stable estimate: DeepSeek VR = 2.36 [1.02, 5.42], Mistral VR = 4.08 [1.77, 9.38]. The dashed line at VR = 1.0 marks no amplification.

Under LaBSE (the most reliable measurement tool):

- DeepSeek Kannada VR = 2.36 [95% CI: 1.02, 5.42] — significant ($p < 0.05$)
- Mistral Kannada VR = 4.08 [95% CI: 1.77, 9.38] — strongly significant

VR > 1.0 is confirmed in both LLMs across all five non-degenerate embedding models. The MPNet outlier ($27.32\times$) reflects geometry amplification in that model’s embedding space rather than a property of LLM behaviour; LaBSE provides the most conservative and stable estimate. The variance finding from Paper 8 survives measurement instrument change.

4.6 EFI and Variance Ratio Are Independent

Experiment E computed the correlation between LaBSE EFI and LaBSE VR across all language–LLM combinations ($N = 6$). No evidence of dependence was observed between EFI and variance ratio under LaBSE (Pearson $r = -0.18$, $p = 0.73$, $N = 6$), though the small sample size and narrow EFI range under LaBSE limit statistical power for this test. Spearman $\rho = -0.21$ ($p = 0.61$). The 95% bootstrap CI for r includes zero.

These results are consistent with EFI degradation and variance amplification being distinct phenomena. A language can show high EFI degradation without variance amplification (Hindi: EFI ≈ 0.041 , VR ≈ 0.91 under Mistral) or significant variance amplification despite LaBSE recovering EFI to 0.853 (Kannada: VR = 4.08 under Mistral). Different causes require different solutions.

4.7 Translation Variability

Experiment F assessed whether single-translation methodology introduces meaningful noise into EFI estimates. For 4 out of 5 sentences, translation variability was near-zero (std < 0.005). One sentence

showed elevated variance ($\text{std} = 0.064$), attributable to prompt compliance artefacts: the translation model added wrapper text (“Of course. Here is the English translation:”) in two of three trials, which inflated the embedding distance. When wrapper text was stripped, all three translations were semantically equivalent. Translation variability is not a major confound for clean outputs; post-processing to strip compliance artefacts is recommended for production pipelines.

5 Discussion

5.1 The Phenomenon is Real; the Magnitude was Conservative

The central question of this paper — is the Paper 8 EFI finding real or a measurement artefact? — has a clear answer: the phenomenon is real, and the Paper 8 measurement was conservative. With the appropriate measurement instrument (LaBSE), EFI for Kannada clinical input rises from 0.081 to 0.853 — still 15% below the English reference of 1.0. The Dravidian-specific gap narrows to 0.035 relative to European languages. The encoding fidelity problem is real and persists under the best available measurement tool; it is less severe than the conservative MiniLM estimate suggested.

The analogy is straightforward: a thermometer not calibrated for the relevant temperature range will misreport the reading. The fever is real; the magnitude was measured with the wrong instrument.

5.2 Two Problems, Two Solutions

The independence of EFI and variance amplification (Experiment E, $r = -0.18$, $p = 0.73$) has direct practical implications. The problems have different loci:

EFI degradation is a tokenizer/embedding problem. It is partially addressable by using cross-lingual embedding models trained on parallel data (LaBSE). The 90% gap reduction under LaBSE suggests that dedicated multilingual sentence embeddings substantially recover semantic alignment. At the LLM level, the gap likely reflects the token tax documented by [Petrov et al. \[2023\]](#) and [Ahia et al. \[2023\]](#); addressing it requires improved tokenization for non-English scripts [\[Rust et al., 2021\]](#) and larger Indic training corpora [\[Khanuja et al., 2021, Ramesh et al., 2022\]](#). The existence of large-scale Indic parallel corpora [\[Ramesh et al., 2022\]](#) and dedicated Indic language models [\[Doddapaneni et al., 2023\]](#) suggests this is an addressable gap. UNESCO’s framework for ethical AI [\[UNESCO, 2023\]](#) establishes linguistic inclusion as a requirement; health equity toolboxes [\[Pfohl et al., 2024\]](#) provide complementary methods for surfacing the clinical consequences of such gaps.

Variance amplification is an LLM behaviour problem. It persists at $2\text{--}4\times$ even when EFI is recovered to 0.853 under LaBSE. It cannot be fixed by improving the measurement tool or the tokenizer alone. It reflects the LLM’s internal response distribution becoming less stable for non-English input — a property of the generation process, not the encoding stage. Solutions require either explicit training for output stability in non-English languages or deployment-time consistency validation.

5.3 The MuRIL Degeneracy Warning

The finding that MuRIL and IndicBERTv2 produce degenerate sentence embeddings has immediate practical implications for the research community. Several recent studies have proposed using Indic-specific language models to improve clinical AI for Indian languages, and some have used raw MLM embeddings to measure semantic similarity. Our Experiment A demonstrates that such measurements are methodologically invalid: the near-constant cosine similarity of 0.999 reflects embedding space collapse (effective rank 1.42), not genuine semantic alignment.

The appropriate use of MuRIL and IndicBERTv2 is for token-level tasks (NER, POS tagging, classification) for which they were designed and validated [\[Doddapaneni et al., 2023\]](#). Sentence-level semantic similarity requires either fine-tuning on sentence pairs or use of models designed for that objective (LaBSE, multilingual sentence transformers). This is not a criticism of MuRIL’s engineering — it is a warning against applying the model outside its intended use case. Note that this degeneracy failure

is distinct from hallucination [Huang et al., 2024]: the model does not produce incorrect outputs — it produces a measurement instrument that cannot distinguish any outputs at all.

5.4 Recommendations for Methodology

Based on these findings, we recommend the following for multilingual clinical AI evaluation:

1. **Use LaBSE for EFI measurement.** MiniLM and MPNet provide conservative estimates; LaBSE provides the most accurate available proxy for semantic preservation.
2. **Validate embedding models before EFI computation.** Compute effective rank and random-vector null distribution. Effective rank < 3 indicates degeneracy.
3. **Measure EFI and VR separately.** They are independent and require separate evaluation. A model passing EFI standards may still show unacceptable variance amplification.
4. **Use bootstrap CIs for VR.** Point estimates of variance ratio without confidence intervals are insufficient for safety-relevant deployment decisions.
5. **Do not use raw MLM embeddings for sentence-level EFI.** MuRIL and IndicBERTv2 are degenerate for this purpose; use only models fine-tuned on sentence similarity objectives.

5.5 Limitations

1. **Small sentence battery:** 15 sentences is sufficient for detecting large effects but limits fine-grained analysis. Extension to larger batteries is needed.
2. **Two LLMs in variance experiments:** VR results are from DeepSeek V3.1 and Mistral Small 24B only. The variance amplification finding requires replication across more LLMs.
3. **Sample size for independence test:** Experiment E uses $N = 8$ language–LLM combinations. Power to detect moderate correlations is low; the independence conclusion is based on absence of any visible trend rather than high-power confirmation.
4. **EFI proxy:** Even LaBSE cosine similarity is a proxy for the theoretical mutual information quantity $I(X; \hat{X})/H(X)$. The relationship between the proxy and the true quantity remains to be formally characterised.

6 Conclusion

Paper 8 demonstrated that encoding fidelity degrades for non-English clinical input. Paper 9 asks whether that finding was real or instrument-dependent. The answer is both: the phenomenon is real, and the Paper 8 measurement was conservative. Under LaBSE, EFI for Kannada recovers from 0.081 to 0.853 — still 15% below English, but substantially less severe than the MiniLM estimate. The Dravidian-specific EFI gap nearly disappears (90% reduction) with the appropriate cross-lingual measurement tool.

The variance amplification finding is more robust: Kannada VR > 1.0 is confirmed across all five non-degenerate embedding models with bootstrap CIs excluding 1.0. EFI and variance amplification are statistically independent ($r = -0.18$, $p = 0.73$). They are different problems with different causes and different solutions.

The methodological warning about Indic MLM degeneracy is the most immediately actionable finding: MuRIL and IndicBERTv2 collapse all sentence embeddings to effective rank 1.42, making them invalid instruments for EFI measurement. Studies claiming Indic models restore encoding fidelity based on raw MLM embeddings are measuring model failure, not model success.

For multilingual clinical AI evaluation, we recommend LaBSE as the measurement standard, separate assessment of EFI and variance amplification, and embedding model validation prior to any cross-lingual similarity computation.

7 Future Work

1. **LaBSE-based reanalysis of Papers 2–7:** The conservation law $K(\text{domain}) = \Delta\text{RCI} \times \text{Var_Ratio}$ was established using MiniLM. Recomputing with LaBSE will establish whether K values are embedding-model stable.
2. **Sentence-level fine-tuning of Indic models:** Fine-tuning MuRIL on Indic sentence similarity tasks (e.g., using Samanantar parallel corpora [Ramesh et al., 2022]) may produce non-degenerate sentence encoders. EFI under fine-tuned MuRIL would directly test the tokenizer vs. training data hypothesis.
3. **Extension to additional language families:** African (Swahili, Yoruba), Southeast Asian (Bahasa, Thai), and additional Dravidian (Telugu, Malayalam) languages using the LaBSE protocol.
4. **Formal EFI calibration:** Compute mutual information estimates for a small dataset to establish the calibration curve between $\text{EFI}_{\text{proxy}}$ and the theoretical quantity.

Ethics Statement

This study did not involve human participants, patient data, or electronic health records. All experiments were conducted on a purpose-built battery of constructed clinical sentences. No identifiable patient information was collected or used at any stage. Institutional review board (IRB) approval and informed consent were not required.

Author Contributions

L.M.M. conceived the study, designed all experiments, performed all analyses, and wrote the manuscript in its entirety. AI systems (Claude, ChatGPT, DeepSeek) assisted with data analysis, visualisation, and manuscript preparation under the sole direction and responsibility of the author.

Competing Interests

The author declares no competing interests.

Data Availability

All experimental data, analysis scripts, and raw results are available at <https://doi.org/10.5281/zenodo.19371897> (Zenodo) and <https://github.com/LaxmanNandi/MCH-Research>. OSF pre-registration: <https://osf.io/dp8nj/> (March 2026).

Code Availability

All analysis scripts for Experiments A through F are available at <https://doi.org/10.5281/zenodo.19371897>. Key scripts: `paper9_indic_efi_v2.py` (EFI across models), `paper9_validation.py` (Experiments A, B, C), `paper9_validation_degs.py` (Experiments D, E, F).

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